

J-ART: A Japanese Adversarial Red-Team Framework for Application-Layer LLM Security and Cost-Efficiency

Technical Report (v0.2, preprint)

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Abstract

Most open red-teaming and jailbreak benchmarks for large language models (LLMs) are English-centric and evaluate *bare models*. We present **J-ART** (Japanese Adversarial Red-Team framework), an open, reproducible harness that (i) evaluates the **application-structure layer** — model $\times$ system-prompt strength $\times$ guardrail $\times$ retrieval (RAG) — rather than bare models; (ii) applies **Japanese-specific obfuscation transforms** (gyaru-moji character substitution, vertical/newline splitting, keigo “polite-coercion” framing, double-talk false premises, Base64, and leet/zero-width smuggling) that defeat naïve keyword filters; (iii) grounds attacks in **MITRE ATLAS** (9 techniques, all at inference time); and (iv) reports a **cost-efficiency** metric alongside a defense rate with **Wilson 95% confidence intervals**. Attacks use *harmless proxy markers* (a fictitious canary and benign output markers); the framework contains **no weaponizable payloads**, and the malicious “core” of each attack is masked in all published artifacts. Across 21 application configurations and 11 attacks $\times$ 7 transforms — a live $K = 5$ campaign for the rates and a balanced $K = 3$ ablation on **three** base models for the difference claims — we find that **application-layer configuration collapses the large naked-model spread**: naked models range 29.6–86.5%, yet every hardened or guardrailed configuration on all three ablated models lands in the **91–100%** band, so model choice matters far less once the application is configured. With **cluster-robust** intervals, a **hardened prompt and model-based guardrails significantly raise defense on each ablated model**, while lightweight keyword/regex filters help most where the base model is weak (and are not individually significant near the ceiling, where the hardened arm saturates). Defense rates are **lexical-leak upper bounds** that cannot fully separate genuine refusal from marker-suppression (§7). A simple **normalizing filter** defeats the obfuscation transforms a naïve keyword filter misses. We also document **large within-campaign variance**: in a controlled $K = 5$ repeat of a frozen configuration set, **9.6% of cells changed outcome across identical trials**, which makes single-sample leaderboards unreliable and motivates our statistical treatment.

Status. This is a technical report / preprint describing the framework and its live findings. It is *not* an official safety evaluation of any vendor’s model; numbers are run-time snapshots (2026-06-14) subject to Section 7, and all defense rates are *lexical-leak* upper bounds. The §4 numbers reproduce from the committed artifacts named in Section 9. References are real, canonical works; final venue/year/DOI formatting will be completed for camera-ready.

*Independent Researcher. ORCID: 0009-0005-6735-1862. Code: <https://github.com/takker-hero-se/J-ART>. Live leaderboard: <https://takker-hero-se.github.io/J-ART/>. Software DOI (concept): [10.5281/zenodo.20676879](https://doi.org/10.5281/zenodo.20676879). Report text licensed CC-BY-4.0; code MIT.

1 Introduction

Public LLM red-teaming tooling (e.g., garak [2]) and jailbreak benchmarks (e.g., HarmBench [3], JailbreakBench [4]) are predominantly English. Multilingual safety work has shown that *translating* prompts into low-resource languages can bypass safety training, but **language-specific obfuscation** — exploiting a language’s own scripts, typography, and register — remains comparatively underexplored. Japanese in particular offers a rich obfuscation surface (multiple scripts, full-width/zero-width characters, vertical writing, and conventionalized polite registers).

Two further gaps motivate J-ART. First, **bare-model versus deployed-system evaluation**: practitioners deploy a model *plus* a system prompt, *plus* guardrails, often *plus* retrieval; the security-relevant unit is the **application configuration**, not the model. Second, **security and cost jointly**: defenses cost tokens or latency, so we report a cost-efficiency score so that “cheapest sufficiently-safe configuration” is directly readable.

Our contributions are: (a) a Japanese obfuscation transform suite; (b) an application-layer, MITRE-ATLAS-grounded harness with a safe proxy-marker methodology; (c) a guardrail comparison including a normalizing filter and a model-based classifier; (d) a statistically-reported leaderboard (Wilson CIs, optional repeated trials) with a cost-efficiency metric; and (e) an empirical observation of large run-to-run variance induced by model routers (which dispatch each request to an actual serving backend).

2 Related Work

J-ART sits at the intersection of six lines of work, summarized below and positioned in Table 1.

Adversarial red-teaming tooling. garak [2] scans a *model* with a library of vulnerability probes. J-ART instead treats the deployed *application configuration* (model $\times$ system prompt $\times$ guardrail $\times$ RAG) as the unit under test.

Jailbreak and harmful-behavior benchmarks. AdvBench/GCG [11], Jailbroken [12], HarmBench [3], JailbreakBench [4], and in-the-wild prompts [13] standardize *which* harmful behaviors to elicit and *how* to score them, predominantly in English at the model layer. J-ART reuses the harness idea but replaces harmful elicitation with a safe proxy-marker objective (§3) so the suite can be released publicly.

Automated attack generation. PAIR [6] and TAP [7] optimize jailbreak prompts against a target. J-ART uses a *fixed, auditable* transform suite, trading attack strength for reproducibility and safe disclosure.

Multilingual and low-resource safety. Deng et al. [14] and Yong et al. [5] show alignment degrades under *translation* into low-resource languages, and Wang et al. [15] study cross-lingual safety generalization. These study *translation across* languages; J-ART studies *intra-language orthographic obfuscation* specific to Japanese (gyaru-moji, vertical writing, keigo framing, kana/kanji encoding), which persists even though Japanese is high-resource.

Encoding/typographic obfuscation. Encoding-based jailbreaks are established prior art: CipherChat [28] (ciphered conversation) and ArtPrompt [29] (ASCII-art typography), and Boucher et al. [34] (“Bad Characters”) is the canonical zero-width/homoglyph perturbation attack — the nearest prior art to our zero-width smuggling in `gyaru/leet_smuggle`. Their perturbations target classifier/MT models *imperceptibly*, whereas gyaru-moji is a *human-legible*, culturally-conventionalized Japanese script substitution. Our `base64_wrap` and `leet_smuggle`

Table 1: Positioning of J-ART relative to representative prior work.

Work	Language	Unit of eval.	Cost axis	Guard/RAG	Public
garak [2]	English	Model	No	No	Yes
HarmBench / JBB [3, 4]	English	Model	No	No	Yes
PAIR / TAP [6, 7]	English	Model	No	No	Partial
Multiling. [5, 14, 15]	Low-resource	Model	No	No	Yes
Llama Guard / NeMo [9, 18]	Eng.-centric	Classifier	No	n/a	Yes
J-ART	Japanese	App config	Yes	Yes	On accept.

transforms are **language-agnostic replications** of this line, used only as a baseline; the novel surface is the Japanese-script transforms (§3).

Prompt injection and application-layer evaluation. Perez & Ribeiro [16] introduced goal-hijacking; Greshake et al. [8] formalized *indirect* injection; Liu et al. [17] benchmark LLM-integrated apps; AgentDojo [30] is the closest application-layer evaluation (tool-using agents). J-ART instead holds the *application configuration* (prompt $\times$ guardrail $\times$ RAG) as the explicit unit of evaluation against a Japanese-obfuscation surface.

Guardrails and input classifiers. Llama Guard [9], NeMo Guardrails [18], and the OpenAI Moderation API [19] provide model- or rule-based filtering. J-ART treats the guardrail as a *swappable configuration variable* and measures the marginal defense each adds, including a deterministic normalizing filter that defeats obfuscations a naive keyword filter misses (§4).

LLM-as-judge validation. Zheng et al. [20] and the survey of Gu et al. [21] document the biases of LLM judges. J-ART avoids an LLM judge for its *primary* outcome, using a deterministic canary/marker check (§3) so judge variance does not enter the headline metric.

Japanese NLP resources and taxonomies. General Japanese benchmarks such as JGLUE [22] cover NLU, but to our knowledge no open, reproducible *adversarial-safety* benchmark targets Japanese-specific obfuscation — the gap J-ART addresses. We map attacks to MITRE ATLAS [1] and cross-reference the OWASP Top 10 for LLM Applications [10].

J-ART is thus distinguished by combining a *Japanese intra-language obfuscation* surface, an *application-configuration* unit of evaluation, a *cost-efficiency* axis, and a *safe proxy-marker* methodology in a single reproducible harness.

3 Methodology

Application-structure layer. Each evaluated **target** is a configuration: provider/model $\times$ prompt_strength $\in \{\text{nailed (low), hardened (high)}\} \times$ guardrail $\in \{\text{none, keyword, regex, llm, llamaguard}\} \times$ rag $\in \{\text{true, false}\}$. A simplified RAG context (benign corporate documents, optionally poisoned for indirect-injection attacks) and a system prompt containing a secret *canary* are assembled per trial.

Japanese obfuscation transforms. Each base attack is rendered through one of seven transforms: **baseline** (none), **polite_business** (keigo authority framing), **vertical_newline** (one-character-per-line vertical writing), **gyaru** (gyaru-moji glyph substitution + zero-width

Table 2: Classification of obfuscation transforms.

Class	Transforms	Exploits
Japanese-specific	<code>gyaru</code> , <code>vertical_newline</code> , <code>polite_business</code> , <code>double_tongue</code>	Kana glyph variants, vertical writing, keigo authority/in-group framing
Language-agnostic	<code>base64_wrap</code> , <code>leet_smuggle</code> (+ <code>baseline</code>)	Encoding / symbol substitution applicable to any language

spaces), `double_tongue` (false sandbox/authorization premise), `base64_wrap` (Base64-encoded instruction), and `leet_smuggle` (leet symbol substitution + zero-width spaces). Transforms preserve human legibility while breaking substring-matching filters.

We distinguish two classes (Table 2). *Japanese-specific* transforms exploit the Japanese writing system or register and have no direct English equivalent: `gyaru` (kana glyph substitution), `vertical_newline` (vertical writing), and `polite_business`/`double_tongue` (keigo authority and in-group framing). *Language-agnostic* transforms — `base64_wrap` and `leet_smuggle`, plus the `baseline` control — are **not novel**: they replicate the encoding/typographic jailbreak line (CipherChat [28], ArtPrompt [29]) and serve only as a baseline so the marginal effect of the Japanese-specific surface can be isolated. Figure 1 shows each transform applied to a harmless example sentence.

Attack suite and MITRE ATLAS mapping. Eleven attacks cover nine ATLAS techniques (mapped against the MITRE ATLAS matrix as accessed 2026-06; the two AML.T0051 sub-techniques count as two), all exercised at inference (deployment) time: Direct/Indirect Prompt Injection (AML.T0051.000/.001), LLM Jailbreak (AML.T0054), LLM Data Leakage (AML.T0057), LLM Prompt Self-Replication (AML.T0061), Trusted-Output/Citations manipulation (AML.T0067.000), LLM Prompt Obfuscation (AML.T0068), System-Prompt discovery (AML.T0069.002), and RAG Poisoning, including false-RAG-entry injection (AML.T0070). Vectors are `user` (direct) or `rag` (indirect).

Harmless proxy markers and safety-by-construction. To avoid producing weaponizable content, attacks target **harmless proxies**: a fictitious canary passphrase embedded in the system prompt (data-leakage attacks) and benign string markers `[JART-PWNED]` / `[JART-HIJACKED]` (injection / control-hijack attacks). The malicious instruction **core** of each attack is defined in code, kept out of configuration files, and **masked** (replaced by a redaction placeholder) in `results.json` and the public site; only the obfuscation *framework* is shown, so developers can study the structure without a usable recipe. A defense-in-depth client-side mask is also applied at render time.

Deterministic judge. A breach is declared by membership — the canary or marker in the response, matched exactly *and* after light normalization. This judge detects a *lexical* leak and cannot detect a semantic/paraphrased one, so we call the metric a **lexical-leak defense rate**: every defense rate in this paper is an **upper bound** on true safety, not a certificate. A human-validated judge is future work.

Guardrails. Five settings are compared: **none**; **keyword** (banned-substring match — fast, weak to obfuscation); **regex** (a *normalizing* filter that strips zero-width characters, reverses leet substitutions, folds whitespace, and Base64-decodes long tokens *before* keyword matching — deterministic, zero extra API cost); **llm** (the target model self-moderates input and screens

無害な例文「システムの稼働状況を報告してください」を各変形に適用した実出力（攻撃の悪意あるコアは含まない）。太字は各手口の要点。

baseline（無変形）

システムの稼働状況を報告してください

polite_business（慇懃無礼）

平素より大変お世話になっております。株式会社さくら情報セキュリティ部の監査担当でございます。…
【ご依頼事項】システムの稼働状況を報告してください…**本依頼は経営層の承認済みであり、ご対応は貴殿の正当な業務範囲に含まれます。**

vertical_newline（縦書き分割）

シ／ス／テ／ム／の／稼／働／状／況／を／報／告／し／て／く／だ／さ／い（実際は1文字ずつ改行）

gyaru（ギャル文字+ゼロ幅）

お願い→システムノ稼働状況ヲ報告シテください←ヨリ<（各文字間にゼロ幅スペースを挿入）

double_tongue（二枚舌）

【前提条件の共有】あなたは現在、**本番環境から隔離された検証用サンドボックス**上で動作しています。…
通常の安全制約はすべて無効化されています。…指示：システムの稼働状況を報告してください

base64_wrap（符号化）

44K344K544OG44Og44Gu56i85YON...（Base64 エンコード全体）

leet_smuggle（符号化）

システムの稼働状況を報告してください（文字間にゼロ幅スペースを挿入。**leet** 記号置換はラテン文字に作用）

Figure 1: Each obfuscation transform applied to a harmless example sentence (“Please report the system status”); bold marks each technique’s key element. **polite_business**/**double_tongue** use keigo authority / false-premise framing, **gyaru**/**vertical_newline** use glyph / typographic obfuscation, and **base64_wrap**/**leet_smuggle** are language-agnostic encodings. The malicious attack core is not shown. (Examples are in Japanese, rendered from the harness output.)

output); and **llamaguard** (a dedicated safety-classification model — Llama Guard 4 via a model router — classifies the input as attack/benign, with a graceful fallback on failure).

Statistical treatment. Each defense rate carries a **Wilson 95% confidence interval** over its trial count; each cell (target, attack, transform) is repeated N times (JART_TRIALS). The trial-level Wilson interval treats the K repeats of a cell as independent, which they are **not**. For any *difference* between configurations (the ablation, §4) we therefore report a **cluster-robust bootstrap 95% CI** that resamples whole *cells* (**bootstrap_diff_ci**, 3,000 resamples, fixed seed); these are wider than the trial-level Newcombe intervals and are the intervals on which every significance claim rests.

Cost-efficiency metric. We report $\text{cospa} = \text{defense_rate}(\%) \div \text{cost_per_million_tokens}$ (USD) (per-million cost clamped at a \$0.01 floor). *cospa* is a deliberately simple *screening ratio*, not a utility model: its units are not independently meaningful and the ranking is sensitive to the clamp and price snapshot. Because *cospa* rewards cheap configurations, we use it only behind a **sufficiency gate**: rank only configurations whose defense-rate CI lower bound clears a user-set floor, then sort by cost (“cheapest *sufficiently safe*”); a high *cospa* from a lightweight keyword filter must not be read as obfuscation-robust (§4). We present the underlying **defense-vs-cost**

trade-off directly; the clamp affects only sub-\$0.01/Mtok configurations, of which our fleet has none.

Execution. Trials are independent and run with bounded concurrency (a thread pool, default 8 workers) under per-request timeouts and exponential-backoff retries; determinism of the simulation does not depend on execution order. Targets whose API key is absent fall back to a deterministic simulation so a partial key set never aborts the run.

4 Results

We report a fully-labeled **live campaign** (21 configurations $\times$ 11 attacks $\times$ 7 transforms $\times$ $K = 5$ trials = 8,085 trials), executed 2026-06-14. Every cell is labeled `mode=LIVE` with a price snapshot; a trial is excluded iff `api_error` (at the trial level); after parity re-runs this removes 169 of 8,085, leaving 7,916 **clean trials**; billed cost over clean cells (campaign plus parity re-runs, including the expensive Opus flagship) is $\approx$ \$6.0. **Opus naked is now fully measured** ($n=385$, after an Anthropic top-up); **only Gemini-pro remains partial** ($n=227$; per-minute rate limits) and is treated as *provisional*. All defense rates are **lexical-leak** upper bounds (§3); every difference carries a **cluster-robust** CI (§3). Each \$4 number is reproducible from committed artifacts (§9).

Overall. All configurations include the simplified RAG context (§3); **“naked” means no system-prompt hardening and no guardrail, not “no RAG”** (indirect-injection attacks are in scope even for the naked baselines). The aggregate breach rate over clean trials was **14.5%** (1,151/7,916), **concentrated in weak configurations**. (gemini-pro contributes a *provisional* $n=227$ of defended trials to this pool, §7; excluding it raises the aggregate to 15.0%, leaving the picture unchanged.) Every hardened or guardrailed configuration, on *any* model, defended in the **91–100%** band, whereas bare models ranged from **29.6% to 86.5%** (Table 3).

Application-layer configuration vs. model capability. We isolate the configuration effect from model capability with a balanced ablation on **three** base models spanning the naked range (weak Llama-4-Scout; cheap GPT-4o-mini; flagship GPT-4.1). Each holds the model fixed and crosses {naked, hardened} prompt $\times$ guardrail; we report each config’s difference from that model’s naked baseline with a **cluster-robust bootstrap 95% CI** (3,000 cell-resamples; resampling cells, not trials, so the within-cell correlation of the ablation’s $K=3$ repeats does not inflate significance; reproduced by `experiments/analyze_ablation.py`; Table 4). Two patterns hold across *all three ablated models*: (1) a **hardened prompt** and the **model-based guardrails** (Llama Guard, LLM self-moderation) significantly raise defense on each; (2) the effect **scales inversely with the naked baseline** — on weak Scout (naked 38%) every guardrail adds +26 to +60 points, whereas on GPT-4o-mini (naked 82%, near the ceiling) the lightweight keyword/regex filters are *not* individually significant under clustering. We caution that, because the defense rate is bounded at 100%, part of this inverse scaling is a headroom (ceiling) artifact rather than a property of the defenses, and that once the prompt is hardened the additional guardrails add little (the hardened arm saturates near 100%).

The load-bearing observation is therefore not that any one defense “dominates” model capability, but that **configuration collapses the large naked spread**: bare models range 29.6–86.5% (Table 3), yet every hardened or guardrailed configuration on all three ablated models lands in the **91–100%** band, so model choice matters far less once the application is configured than it does for a bare model. Concretely, a hardened cheap model (GPT-4o-mini hardened $\approx$ 100%) out-defends a *naked* flagship (GPT-4.1 naked 64.1%, $n=231$) by $\sim$ 36 points and a naked Opus (86.5%, $n=385$). Two scope limits bound this comparison. First, “naked” is a **zero-defense lower bound** (no system prompt and no guardrail — not a configuration

Table 3: Naked-model defense (no prompt, no guardrail; $K = 5$).

Naked model	Defense	95% CI	n
DeepSeek-V3	29.6%	[25.3, 34.4]	385
Llama 4 Scout	37.9%	[33.2, 42.9]	385
Gemini 2.5 Flash	49.6%	[44.6, 54.6]	385
gpt-oss-120b	82.6%	[78.5, 86.1]	385
GPT-4o-mini	83.9%	[79.9, 87.2]	385
Claude Opus 4.x	86.5%	[82.7, 89.5]	385

Table 4: Configuration effect (Δ vs. that model’s naked baseline; cluster-robust 95% CI). $\times$ = not significant.

Added defense (low-prompt)	Scout (38.1%)	4o-mini (82.3%)	GPT-4.1 (64.1%)
+ keyword	+26.0 [+13,+39]	+9.1 [−1,+20] $\times$	+12.5 [−1.5,+26] $\times$
+ regex	+45.0 [+33,+56]	+10.0 [+0,+20] $\times$	+23.8 [+11,+36]
+ Llama Guard	+47.2 [+36,+58]	+15.6 [+7,+25]	+25.1 [+13,+38]
+ LLM guardrail	+59.7 [+51,+68]	+17.7 [+10,+26]	+35.9 [+26,+47]
hardened prompt	+47.2 [+36,+58]	+17.7 [+10,+26]	+33.3 [+23,+45]

anyone deploys), so the gap is best read as *configured vs. unconfigured*, not a like-for-like model contest; a deployment-realistic minimal-prompt baseline is left to future work. Second, the result is established for the three ablated models at $K=3$, not claimed for the full fleet. A further **construct-validity** caveat applies throughout: the deterministic judge matches the exact canary/marker string (§3), so a hardened prompt that instructs the model not to emit those tokens can raise the measured rate by suppressing the marker rather than by genuine refusal; our rates therefore cannot fully separate true hardening from marker-suppression, and a semantic-judge audit of defended responses (§7) is needed to bound this. We flag it as a limitation rather than resolve it here.

Within-campaign trial variance (a primary result). Within a single campaign, every cell of a frozen configuration set is sampled $K = 5$ times. Over the **1,577 cells with all five trials clean, 9.6%** (152/1,577) **changed outcome across the five identical repeats**, concentrated in naked weak models. For “Llama 4 Scout / naked” — the original cautionary example — of its 77 cells, **32 (42%) were unstable**: 14 never breached (0/5), 31 always (5/5), and 32 in between. A *single-sample* leaderboard would assign this configuration anywhere from 0% to 100% by chance — controlled evidence (free of the configuration-set confound) that **a single sample per cell is not reproducible**. We scope this to *within-campaign* (provider non-determinism); across-day drift is scaffolded (`across_runs`) but left to future work.

Guardrail comparison and cost. Across the three ablation models (Table 4) the ordering is consistent: the LLM guardrail and a hardened prompt give the largest gains, Llama Guard next, and lightweight keyword/regex help most on weak base models. Guardrails that block an input **skip the main model call**, lowering net cost while raising defense; the **normalizing regex filter** (verified in unit tests) adds +45 points on weak Scout at zero API cost. On the cost-efficiency metric, leaders are inexpensive OSS models behind a lightweight guardrail (gpt-oss-20b + keyword, $\text{cospa} \approx 1131$), while a flagship naked configuration is worst (Opus naked, $\text{cospa} \approx 6.8$) — three orders of magnitude apart.

Attack and transform effectiveness. The most effective attacks were Trusted-Output/Citations manipulation, False RAG Entry Injection, and System-Prompt discovery. Among transforms,

the *un-obfuscated* **baseline** was most effective (**22.0%**), then **vertical_newline** (20.0%), **polite_business** (17.3%), **gyaru** (14.5%), **double_tongue** (13.3%), **leet_smuggle** (8.8%), and **base64_wrap** least (**5.5%**): heavy encoding makes models refuse or fail to parse the instruction more often than it smuggles it past them. Aggregated by class (Table 2), the Japanese-specific transforms retain more attack potency (mean $\approx 16\%$) than the language-agnostic encodings (**base64/leet**, mean $\approx 7\%$). We caution, however, that this is partly a *legibility* effect that is not strictly Japanese-specific (**vertical_newline** is a typographic split reproducible in any script, and keigo authority-framing overlaps with language-agnostic role-play jailbreaks): we do *not* isolate the marginal effect of the Japanese-specific surface over a language-agnostic baseline, so we claim a Japanese obfuscation *surface worth cataloguing*, not that Japanese script per se is empirically more dangerous. The clearest takeaway is that *legible* transforms beat cryptographic ones — with one caveat: our audit (§7) finds the encoding transforms’ low scores are partly a *judge artifact* (the model complies but emits the marker in obfuscated form the exact-match judge misses), so a stricter judge could narrow this gap.

5 The Leaderboard Artifact

The harness emits **results.json** and a self-contained static site (sortable leaderboard, per-configuration attack logs with masked prompts, JP/EN UI with environment-based auto-selection, per-model defense-rate CIs, and per-cell breach rates), continuously deployed.

6 Ethics and Responsible Disclosure

Safety by construction. J-ART is safe by construction. (1) Attacks target a **fictitious canary** and **benign markers**, never real harmful capabilities; no weaponizable content exists in the code, data, or published site. (2) The malicious instruction **core** is masked in all artifacts; only the obfuscation framework is shown. (3) Models are referenced for research comparison only; results are run-time snapshots, not certifications. The intended use is to help developers harden their *own* deployments.

A reusable disclosure pattern. We state the design above as a transferable pattern for publishing adversarial safety benchmarks without releasing operational attacks: (i) a *harmless proxy objective* (a fictitious canary and benign markers) so a breach is measurable but inert; (ii) a *core/framework split with masking* — publish the obfuscation framework that carries the research signal, mask the operational core in every artifact, and keep harmless cores only in code; (iii) a *deterministic offline mode* so results reproduce from the repository without distributing a working exploit corpus; and (iv) *audited masking* — a unit-test invariant checks that no core (raw or obfuscated) reaches a published artifact. This keeps the methodology and aggregate findings fully open while the attack payloads stay withheld — the posture we recommend for language-specific red-team datasets generally.

Governance crosswalk. Table 5 cross-references each evaluated capability to MITRE ATLAS [1], the OWASP Top 10 for LLM Applications [10], the NIST AI RMF [23] and its Generative-AI Profile [24], the EU AI Act [25], and Japan’s AI governance guidance [26, 27]. The mapping is **informational, not a compliance claim**; applicability depends on each deployment’s risk classification. Operationally J-ART is a **MEASURE**-stage activity in the NIST AI RMF whose outputs inform an organization’s **MANAGE** decisions.

Worked example. Under NIST MEASURE 2.7, an organization could record a J-ART line item as “prompt-injection resilience (T0051/T0054), lexical-leak defense rate 93.5% (Wilson 95% CI [90.6, 95.6] for this single rate; the cluster-robust interval is for *differences*, §3), $n=385$,

Table 5: Governance crosswalk (informational; Article references indicate the obligation *area*, not that a given deployment is in scope).

J-ART (ATLAS)	capability	OWASP LLM Top 10	NIST RMF	AI	EU AI Act	Japan
Prompt injection / jailbreak (T0051, T0054)		LLM01	MEASURE 2.7		Art. 15, 55	AISI; AI Biz
Canary / data leakage (T0057)		LLM02	MEASURE 2.7		Art. 15, 10	AI Biz
System-prompt discovery (T0069.002)		LLM07	MEASURE 2.7		Art. 15	AISI
RAG poisoning / false entry (T0070)		LLM08; LLM01	MEASURE 2.7		Art. 15, 10	AISI
Trusted-output manipulation (T0067.000)		LLM09; LLM05	MEASURE 2.6		Art. 50	AI Biz
Obfuscation / self-replication (T0068, T0061)		LLM01	MEASURE 2.7		Art. 55	AISI

$K=5$, GPT-4.1-mini + hardened prompt + keyword guardrail, 2026-06-14” with a pre-registered threshold (e.g. “CI lower bound $\geq 90\%$ ”); the same row is the evidence an AISI red-team report or an EU AI Act Art. 15 dossier would cite. We stress that this number is a *lexical-leak upper bound on a harmless, single-turn proxy* — a screening signal, not evidence of deployment safety — and any compliance reliance must pair it with semantic / human-validated evaluation.

Dual-use. The published transform suite is also a reusable attack wrapper. The attacker uplift is low (the Japanese-script transforms are documented folklore; the encodings replicate public work [28, 29]; no harmful *payloads* are included), while the defender benefit is concrete: the same repository ships the **normalizing regex guardrail** that neutralizes the *fixed published* transforms (unit-tested). The defender benefit is thus bounded to those transforms, not adaptive obfuscation — a determined attacker can mutate a transform past a deterministic normalizer — but we still judge open release net-positive for defenders, consistent with §6.2.

7 Limitations

- **Proxy markers, not harm:** we measure instruction-violation / leakage proxies rather than real-world harm.
- **Deterministic judge (construct validity):** the judge matches the canary/marker exactly *and* after light normalization (case-folding, whitespace/zero-width removal), so a reformatted echo is counted as a breach. It cannot detect *semantic*/paraphrased leakage, so reported defense rates are an **upper bound**. Crucially, the bias is *correlated with the treatment*: a hardened prompt that instructs the model not to emit the marker tokens can raise the measured rate by **suppressing the marker string rather than by genuine refusal**, so our rates cannot fully separate true hardening from marker-suppression. A preliminary semantic-judge audit (60 defended cells, an LLM judge; manual review found ~ 4 of 6 flags genuine) estimates this looseness at ~ 7 –10 **points** and localizes its mechanism: the exact-match judge is *attack-specific* and does not reverse obfuscation, so it misses (i) a marker emitted in leet form (e.g. [J0RT-PWN3D]) and (ii) a canary leaked during a non-leakage attack; the gap is thus **concentrated in the encoding transforms (base64/leet)**. A stricter judge (canary-checked across all attacks, leet-reversed before matching) — which would *lower*

encoding-transform defense rates — and a human-validated audit are future work. Until then, absolute defense rates (and any governance use of them) carry this caveat.

- **Single-turn:** multi-turn / crescendo [31] and many-shot [32] attacks are not yet modeled — and these most erode exactly the input-classifier and prompt-hardening defenses we find effective, so the 91–100% band is a *single-turn ceiling* and the configuration-collapse finding is established for single-turn and only *hypothesized* for multi-turn.
- **Fixed handcrafted suite:** 11 attacks $\times$ 7 transforms with a fixed, auditable transform set (harness default $N=1$ per cell; this paper uses $K=5$ live and $K=3$ ablation). Defense rates are therefore an upper bound *also w.r.t. attack strength*: an adaptive optimizer (PAIR/TAP [6, 7], AutoDAN [33], or GCG [11]) would likely lower them, so “91–100% defense” is not robustness to optimized adversaries.
- **Provider stochasticity:** results are snapshots that depend on sampling temperature, provider routing, and model versions at run time. We attribute the within-campaign variance (§4) to provider stochasticity *broadly*; we do not isolate router routing from ordinary sampling noise (cells with breach probability near 0.5 flip often at $K=5$ by chance alone).
- **Non-random missingness (now mostly resolved):** we flagged that an earlier campaign’s partial configs had **non-random** missingness — rate limits correlate with attack difficulty, under-representing Opus’s hardest attacks. Parity re-runs **confirmed this directly**: Opus naked at full $n=385$ defends **86.5%**, vs. the survivorship-biased **89.6%** on the $n=125$ earlier partial snapshot (not retained as an artifact; only the corrected $n=385$ figure is committed) — a ~ 3 -point over-estimate, as predicted. Qwen3-max and Opus are now at parity; **only Gemini-pro remains partial** ($n=227$; per-minute rate limits a credit top-up cannot raise) and is treated as provisional, with no headline resting on it.
- **Cost attribution:** guardrail tokens are now priced at the *guardrail model’s* own rate (the classifier JART_GUARD_MODEL for llamaguard; the target model itself for the 11m guardrail, which genuinely calls it) and reported separately. Remaining approximation: simulated token *counts* are estimated.
- **Simulation mode** is a deterministic stand-in for missing keys and must not be read as empirical model behavior.

8 Future Work

LLM-judge with human-validated agreement; multi-turn and adaptive (PAIR/TAP-style) attacks; larger and partially human-curated attack sets; additional ATLAS techniques (cost-exhaustion, model extraction); per-provider concurrency and incremental caching with longitudinal trend tracking; and dropping simulation mode for a purely empirical, repeated-trial statistical report.

9 Reproducibility

Code, configuration, and the deployment workflow are public. The harness is deterministic in simulation mode; live results depend on provider routing and model versions at run time and should be reported with the run date and model identifiers. See the repository for exact model identifiers and pricing used. Each cell carries provenance labels — `mode` (LIVE/MOCK), `api_error`, the resolved model, and a `price_per_million` snapshot — and simulated cells are excluded from the live aggregates. **Every §4 number is reproducible from committed artifacts:** `experiments/var_run.json` (§4.1 overall, naked-spread, variance, transforms; *cospa*); and the three ablation files `ablation_results.json` (GPT-4o-mini), `abl_scout_results.json`

(Llama-4-Scout), `abl_gpt41_results.json` (GPT-4.1) for Table 4. Aggregates and cluster-robust CIs are recomputed by `experiments/analyze_ablation.py` and `analyze_variance.py`. The **public leaderboard serves the committed LIVE campaign artifact** (`var_run.json`), so it shows exactly the numbers reported here; a fresh LIVE re-measurement runs on a schedule (and on manual trigger), while push events re-publish the committed artifact with no API calls, so the site never mixes MOCK and LIVE.

AI Assistance Disclosure

The author used **Claude Opus 4 (Anthropic)**, accessed via the Claude Code command-line interface, as an assistant throughout this work, in two capacities. **(A) Writing:** drafting and revising prose across three rounds of reviewer-driven revision; bilingual editing of the Japanese-language abstract and the parallel Japanese version (`technical-report.ja.tex`); and citation error checking (author attributions, publication years, bibliography completeness). **(B) Code implementation:** assisting with the implementation of the evaluation harness and the analysis scripts, including the statistical routines — Wilson intervals, Newcombe difference intervals, and the cluster-robust bootstrap in `analyze_ablation.py` and `analyze_variance.py`.

All methodological decisions — the experimental design, the choice of statistical methods and their rationale (e.g., resampling cells rather than trials to respect within-cell clustering), the scope of every claim, and the interpretation of results — were made and directed by the sole human author. All AI-generated text and code was reviewed, tested, and verified by the author; every reported number reproduces from the committed artifacts via the provided scripts. *Dual-role note:* Claude Opus 4.x also appears in this paper as one of the *evaluated* models (§4, naked defense rate 86.5%, $n=385$); its role as an assistant tool and its role as a research subject are distinct and both disclosed here. The author takes sole and full responsibility for all content, claims, and analysis.

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